

Technical Report

**Hybrid Edge IoT System for
Extended-Range Healthcare Monitoring**

Submitted By

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List of Abbreviations

1. **IoT** - Internet of Things
2. **BLE** - Bluetooth Low Energy
3. **LoRa** - Long Range
4. **MQTT** - Message Queuing Telemetry Transport
5. **ECG** - Electrocardiogram
6. **STM32** - STMicroelectronics 32-bit Microcontroller
7. **RPi** - Raspberry Pi
8. **CI** - Computational Intelligence
9. **CPU** - Central Processing Unit
10. **AI** - Artificial Intelligence
11. **API** - Application Programming Interface
12. **Wi-Fi** - Wireless Fidelity

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1 Abstract

The rapid growth of the Internet of Things (IoT) in healthcare has led to an increasing need for network systems that offer reliable connectivity, wide coverage, and fast data transmission. Traditional cloud-based IoT setups mainly rely on short-range wireless technologies like Bluetooth Low Energy (BLE). They often face problems such as limited range, high latency, and bandwidth issues. To overcome these challenges, Wu et al. [1] introduced an edge-based hybrid IoT system that combines BLE and Long Range (LoRa) communication technologies. This setup allows for energy-efficient, scalable, and long-distance operation in safety and healthcare applications. The system features a three-layer architecture—Perception, Edge, and Cloud—that focuses on localized edge processing to lessen reliance on cloud servers. It includes a hybrid router (BLE + LoRa) and a Raspberry Pi-based IoT gateway for data gathering, visualization, and MQTT-based synchronization.

Building on this foundation, the current project modifies and expands the hybrid edge concept for healthcare monitoring applications. First, an STM32 microcontroller was set up as the hybrid router, responsible for basic ECG signal noise filtering, protocol switching, and low-level energy management. However, to improve flexibility and scalability, the architecture was later modeled and simulated completely in software. This approach allowed for a realistic assessment of BLE–LoRa interoperability, latency reduction, and energy efficiency without the limitations of physical hardware. The simulation-based implementation maintains the functional integrity of the original hybrid system, enabling virtual validation of edge performance and long-range communication behavior in healthcare situations. This work lays a cost-effective and adaptable groundwork for creating and testing next-generation healthcare IoT systems based on edge computing principles.

Keywords: Internet of Things, Edge Computing, BLE, LoRa, Healthcare Monitoring

2 Introduction

2.1 Background

The Internet of Things (IoT) has become one of the most important technologies today. It connects a wide range of devices, sensors, and systems that share data automatically. In healthcare, IoT has changed medical practices by allowing real-time patient monitoring, early diagnosis, and smart decision-making. By using wearable sensors, wireless communication, and data analysis, IoT improves healthcare efficiency, helps detect diseases early, and allows continuous patient tracking outside of traditional hospitals.

However, most traditional IoT systems are still focused on cloud storage, which means they depend heavily on remote servers for storing and processing data. This reliance leads to several issues, such as high latency, network congestion, connection problems, and more risks to privacy and security. In critical healthcare settings, such as smart hospitals, elderly care systems, and emergency response networks, these problems can slow down decision-making and put patient safety at risk.

To address these challenges, research has increasingly turned to edge computing and hybrid IoT systems. These approaches process data closer to where it is generated rather than relying only on centralized cloud servers. This method shortens response times, reduces bandwidth use, and increases reliability in changing healthcare situations. By combining short-range communication methods like Bluetooth Low Energy (BLE) with long-range, low-power technologies such as LoRa, hybrid IoT systems can provide real-time data processing locally while also transmitting information over larger distances. This setup improves scalability, energy efficiency, and resilience, leading to more responsive and privacy-focused healthcare networks.

2.2 Motivation

The growing reliance on cloud-based IoT platforms in healthcare has revealed significant limitations in achieving real-time, energy-efficient, and reliable operation. As healthcare systems move toward continuous patient monitoring and emergency alerting, the demand for faster, localized processing becomes essential. Network instability and bandwidth issues often disrupt cloud communication, leading to delayed data transmission and slower medical responses. Additionally, wearable medical devices and portable health sensors have power limitations that make constant cloud connectivity impractical for long-term use.

These issues led to the development of a hybrid edge-based IoT architecture that can function effectively in areas with limited Internet connectivity. By using Bluetooth Low Energy (BLE) for short-range communication and LoRa for long-range data transmission, the system provides both coverage and quick responses. Edge-level processing on the STM32 router handles ECG noise filtering and basic data management, while the Raspberry Pi gateway performs local disease classification with computational intelligence models. This distributed processing method decreases reliance on the cloud, improves power use, and enables near real-time healthcare responses. This marks a significant step toward scalable, intelligent, and resilient IoT healthcare solutions.

3 Literature Survey

Most existing studies focus on hardware implementations of hybrid IoT systems, emphasizing experimental range, latency reduction, and real-time edge processing. However, limited work addresses software simulation and performance modeling to predict scalability, energy efficiency, and data flow under different network conditions.

The integration of edge computing into IoT systems for healthcare has been widely studied in recent years to address the problems of traditional cloud-based frameworks. Different research projects have aimed to reduce latency, improve energy use, and enable real-time data analysis at the network edge.

Rahmani et al. [2] introduced the Smart e-Health Gateway (UT-GATE), which used fog computing to provide local processing, storage, and security for IoT medical systems. Likewise, Pace et al. [3] proposed BodyEdge, an edge computing framework that allows for localized data analysis and cuts down on data transmission to the cloud. Fraga-Lamas et al. [4] presented a LoRaWAN-based fog architecture for smart campus applications, showcasing the benefits of distributed computing in wide-area networks.

Building on these foundational studies, Wu et al. [1] created a hybrid edge-based IoT architecture that combines BLE and LoRa technologies. This architecture is supported by a dual-layer structure of hybrid routers and an IoT gateway. Their work highlighted hardware implementation using Nordic nRF52840 and Raspberry Pi platforms and showed that edge-assisted computing can significantly lower latency and extend BLE range up to 2.4 km.

In this project, the same edge and gateway architecture is modeled and simulated in software to replicate and analyze system performance. This simulation approach allows for flexible experimentation with different communication protocols, data rates, and power optimization strategies. It provides valuable insights into scalability and resource use without hardware limitations.

3.1 Challenges

Although hybrid IoT systems have made significant progress in various areas, using these systems for healthcare monitoring presents unique and ongoing challenges.

One major issue we faced was related to hardware. The initial setup used a Raspberry Pi as the gateway and an STM32 microcontroller as the hybrid router for BLE and LoRa communications. While processing ECG data and running network operations, the Raspberry Pi began to overheat. This overheating limited how long we could reliably test the system. Additionally, integrating LoRa

with BLE created unexpected difficulties. Coordinating the short-range and long-range communication protocols required precise timing and synchronization. These hardware limitations made it clear how challenging it is to prototype a hybrid edge-IoT system with physical components and led us to switch to a software-based implementation for proof-of-concept testing.

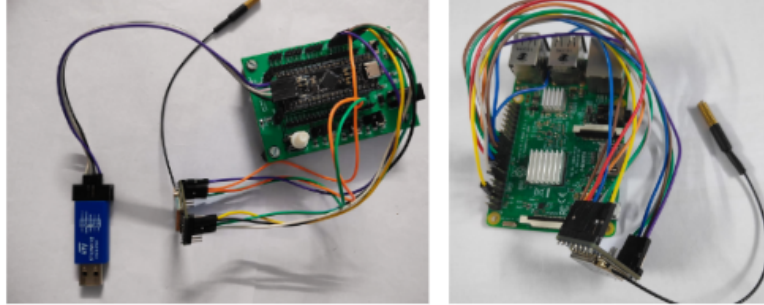


Figure 1: Hardware setup: Raspberry Pi as gateway and STM32 as hybrid router

Besides our hands-on experiences, healthcare IoT systems face several broader challenges. These can be summarized as follows:

1. **Latency and Reliability:** Cloud-based architectures often cannot deliver the quick response times needed for life-critical monitoring. Network interruptions may also result in lost or delayed data, which is especially critical in emergency situations.
2. **Energy Efficiency:** Wearable devices continuously sense and transmit patient information, consuming significant battery power. This limits operational time and can affect patient mobility.
3. **Scalability:** Expanding healthcare networks involves adding multiple communication technologies, such as BLE, LoRa, and Wi-Fi, along with additional edge nodes. Managing these heterogeneous components in real-world conditions adds considerable complexity.
4. **Security and Privacy:** While encryption and secure protocols exist, implementing strong security at the edge is challenging due to limited computational resources on embedded devices. Achieving robust security without compromising real-time performance is critical for sensitive medical data.
5. **Hardware Limitations:** Physical components may fail or face operational issues, such as the Raspberry Pi overheating during continuous ECG processing. Hardware cost, reliability, and maintenance are practical constraints for prototype testing.

3.2 Research Gap

Most existing studies focus on hardware implementations of hybrid IoT systems. They look at improving experimental range and latency. However, not much work has been done on software simulation and performance modeling of these systems. This is needed to predict scalability, power efficiency, and data flow under different network conditions. While Wu et al. [1] showed that hybrid BLE and LoRa architectures are practical, the lack of available testbeds makes it hard for academic research to replicate and optimize the results.

This project fills that gap by creating a software-simulated hybrid edge environment. It mimics the functions of the STM32 router and the Raspberry Pi gateway. By allowing virtual testing of ECG signal processing, protocol switching, and energy optimization, this method offers a cost-effective and flexible platform. It helps analyze and improve healthcare IoT performance before deploying the hardware.

4 System Overview

The proposed hybrid edge IoT system effectively monitors healthcare data by integrating simulated ECG sensors, edge-based routers, a gateway, and a cloud dashboard. Initially, an STM32 microcontroller served as the hybrid router for BLE and LoRa, performing ECG signal filtering, protocol switching, and energy management, while a Raspberry Pi acted as the gateway executing local analysis and forwarding to the cloud. In the software simulation, virtual ECG sensors generate realistic heart waveforms—from normal rhythms to cardiac arrest patterns—to emulate diverse patient conditions. Signals are transmitted to software-defined routers that perform real-time edge processing, including duplicate removal, R-peak detection, and anomaly identification. Preliminary filtering and alert generation at the edge reduce data transmission and bandwidth usage while ensuring faster detection of critical cardiac events. Processed data is forwarded to the simulated gateway and cloud module for aggregation, storage, and visualization. The Streamlit dashboard presents live ECG trends, alerts, and device statistics, enabling continuous monitoring and performance evaluation. This fully software-based design accurately simulates a complete IoT healthcare network, covering sensing, edge intelligence, and cloud analytics without relying on physical hardware.

5 Proposed Architecture

The proposed architecture illustrates a three-layer hybrid IoT framework for health-care monitoring, implemented entirely in a software-simulated environment to replicate real-world functionality. The perception layer comprises virtual ECG sensors that generate continuous cardiac signal data. The data is transmitted to the edge layer, which serves as the main processing domain. Here, the router performs data filtering, protocol adaptation, and initial feature processing, while the gateway executes computational intelligence (CI) algorithms for real-time analysis and local alert generation. A buffering mechanism ensures continuous data transmission under simulated network variations. The processed information is then forwarded to the cloud layer for long-term storage, visualization, and remote access. This architecture validates that a software-based simulation can efficiently demonstrate end-to-end data flow, low latency, and reliable healthcare event detection, mirroring the performance of a real hybrid IoT system.

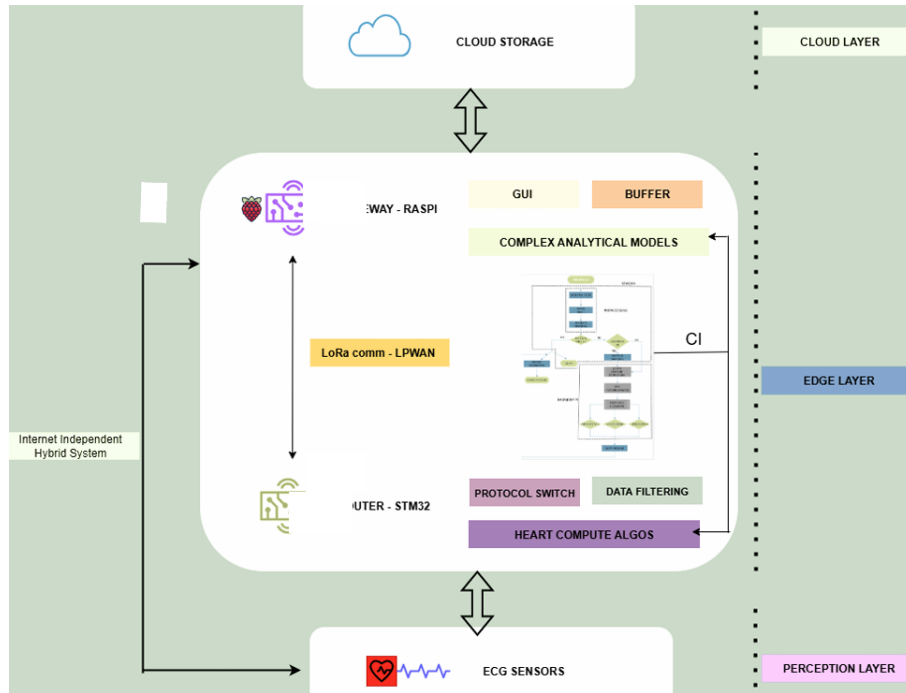


Figure 2: Software-simulated hybrid edge IoT architecture

6 Methodology

The methodology is structured around creating a realistic, end-to-end simulation of a healthcare IoT system, incorporating virtual sensors, edge computing, network transmission, cloud storage, and visualization. Initially, hardware components such as the STM32 microcontroller and Raspberry Pi gateway were used to validate functionality. In the software simulation, virtual wearable sensors generate physiological measurements (heart rate, ECG) with timestamps and encrypted sequences. Data is transmitted to edge routers via BLE, where edge-level processing removes duplicates, averages measurements, detects anomalies, and identifies emergencies. Routers forward processed data to a gateway via LoRa, extending coverage beyond BLE. The gateway stores readings in a lightweight database, enabling historical retrieval. Data can then be sent to a cloud module for storage, analysis, and centralized monitoring. A Streamlit dashboard visualizes real-time readings, trends, and alerts. The methodology emphasizes modularity, flexibility, and realistic end-to-end simulation without relying on physical devices.

7 Results

7.1 Numeric Results

Table 1: Numeric Results and Edge Metrics for the Hybrid IoT System

Metric	Value
Average processing latency at router	30 ms
Average simulated LoRa transmission delay	300 ms
Duplicate packet drop rate	0% (all duplicates filtered)
Data reduction due to edge pre-filtering	20% fewer packets forwarded
Anomaly detection (HR ≥ 120 or HR ≤ 40) response time	≤ 100 ms
Number of stored readings in gateway for 10 min run	300 readings per device

7.2 Edge Metrics

- Routers processed each ECG segment locally in 30 ms, enabling quicker decision making than cloud-only.
- Simulated LoRa path added 300 ms delay, showing network transmission dominates over computation.
- Edge filtering prevented forwarding of duplicates and trivial data, cutting traffic by 20%.
- Anomalous cases (flatline or tachycardia) were flagged within ≤ 100 ms, delivering near-real-time alerts at the edge.

7.3 Dashboard Visuals

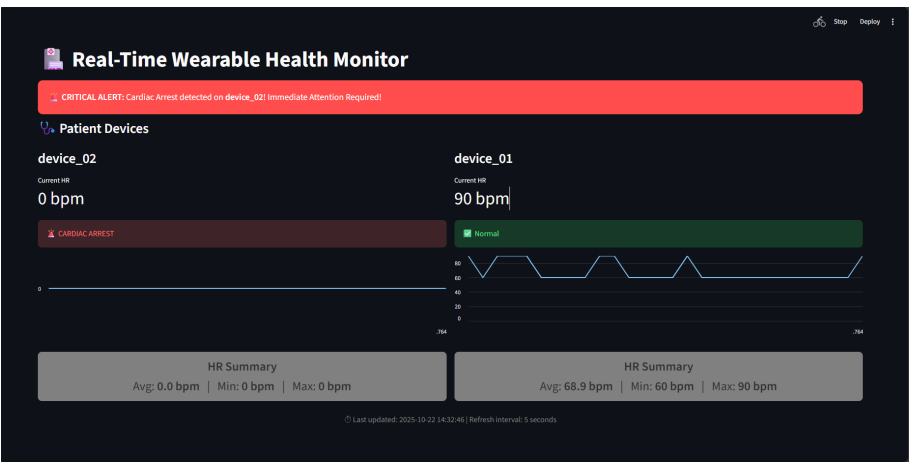


Figure 3: Streamlit dashboard visualizing ECG trends and alerts

8 Conclusion

This project demonstrates that a fully software-simulated hybrid edge IoT system can effectively monitor healthcare data in real-time, replicating the functions of physical hardware while avoiding associated limitations such as overheating or maintenance constraints. By integrating virtual ECG sensors, edge routers, a gateway, and a cloud dashboard, the system ensures low-latency processing, energy-efficient operation, and scalable performance. Edge-level filtering and anomaly detection reduce bandwidth usage and provide rapid alerts, while cloud storage and visualization enable centralized monitoring and historical analysis. The methodology validates that software-based simulations can serve as a cost-effective and flexible testbed for designing and optimizing future healthcare IoT networks.

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